

ROMY RODRÍGUEZ MARTÍNEZ

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RESEARCH INTERESTS

I am broadly interested in the detection and characterization of low-mass extrasolar planets. I use ground- and space-based observations of exoplanets and their host stars to determine their fundamental properties and constrain the interior structure of small planets. I am also interested in the formation and evolution of M-dwarf planetary systems and exoplanet demographics, and in the links between the composition of terrestrial planets and the chemical composition of their host stars.

EDUCATION

Ph.D., Astronomy The Ohio State University, Columbus, OH, USA Advisor: B. Scott Gaudi	June 2023
Master of Science, Astronomy The Ohio State University, Columbus, OH, USA	May 2020
Master of Science, Physics University of Puerto Rico, Río Piedras Advisor: Sarah Ballard	June 2017
Bachelor of Science, Physics University of Puerto Rico, Río Piedras	August 2009 - May 2015

APPOINTMENTS

Future Faculty Leaders Fellow Harvard University Center for Astrophysics, Harvard & Smithsonian, MA, USA.	2023–Present
Astronomy Graduate Student The Ohio State University, OH, USA.	2017–2023
Physics Master Student University of Puerto Rico, Río Piedras, PR, USA.	2015–2017

PUBLICATIONS

24 publications in total: 10 First Author, 14 Contributing Author

[First Author Publications](#)

10. **Rodríguez Martínez, R.**, “The Relationship between Rocky Planet Composition and Host Star $[\alpha/\text{Fe}]$ ”, *under review*
9. **Rodríguez Martínez, R.**, E.K. Pass, Cargile, P.A., et al. “A Uniform Determination of the Bulk Metallicities and Alpha Enrichments of Confirmed Exoplanet Systems with TRES”, *The Astrophysical Journal*, 1004, 91. [arXiv:2605.11075](#)
8. **Rodríguez Martínez, R.**, Eastman, J.D., Collins, K.A., et al. “Discovery and Characterization of an Eccentric, Warm Saturn Transiting the Solar Analog TOI-4994”, *The Astronomical Journal*, 169, 72. [arXiv:2412.02769](#)

7. **Rodríguez Martínez, R.**, Gaudi, B.S., Martin, D.V., et al. “A Comparison of the Composition of Planets in Single- and Multi- Planet Systems Orbiting M dwarfs”, 2023, *The Astronomical Journal*, 166, 137. arXiv:2307.13034
6. **Rodríguez Martínez, R.**, Gaudi, B.S., Schulze, J., et al. “A Reanalysis of the Composition of K2-106b: an Ultra-short Period Super-Mercury Candidate”, 2023, *The Astronomical Journal*, 165, 97. arXiv:2208.07883
5. **Rodríguez Martínez, R.**, Stevens, D.J., Gaudi, B.S., Panero, W., “Analytic Estimates of the Achievable Precision on the Physical Properties of Transiting Planets using Purely Empirical Measurements”, 2021, *The Astrophysical Journal*, 911, 84. arXiv:2101.09289
4. **Rodríguez Martínez, R.**, Gaudi, B.S., Rodriguez, Joseph E., “KELT-25b and KELT-26b, a Hot Jupiter and a Substellar Companion Transiting Young A-stars observed by TESS”, 2020, *The Astronomical Journal*, 160, 3. arXiv:1912.01017
3. **Rodríguez Martínez, R.**, Lopez, L., Shappee, B., Kochanek, C.S., Jayasinghe, T., Schmidt, S., “A Catalog of Flares on M dwarfs with ASAS-SN”, 2020 *The Astrophysical Journal*, 892, 144. arXiv:1912.05549
2. **Rodríguez Martínez, R.**, Ballard, S., Mayo, A., Vanderburg, A., Montet, B., Christiansen, J. L., “Characterization of Low Mass K2 Planet Hosts Using Near-Infrared Spectroscopy”, 2019, *The Astronomical Journal*, 158, 155. arXiv:1808.03652
1. **Rodríguez Martínez, R.**, Schmidt, S. J., Jayasinghe, T., Stanek, K. Z., Prieto, J. L., Shappee, B., Kochanek, C.S., Thompson, T. A., Holoién, T. W.-S., Bersier, D., Brimacombe, J., “ASASSN-18di: Discovery of a Powerful Flare on a Mid-M Dwarf”, 2018, *Research Notes of the American Astronomical Society*, 2, 2. arXiv:1804.04673

Contributing Author Publications

14. Behrard, A., Sagar, S., Yuxi, L., **Rodríguez Martínez, R.** et al., “M Dwarf Singles and Multis Have Dissimilar Metallicities Indicating Different Origins”, *under review*, arXiv:2609.27013
13. DiTomasso, V., Yee S.W., **Rodríguez Martínez, R.**, et al. “A Sample of Thick-Disk Stars Hosting Hot Jupiters”, *Under review*
12. Blunt, S., Nielsen, E., Newton, E., et al. (incl. **Rodríguez Martínez, R.**), “A statistical method for constraining the capability of the Habitable Worlds Observatory to understand ozone onset time in Earth analogs” *Journal of Astronomical Telescopes, Instruments, and Systems*, Volume 11, id. 042214. arXiv:2507.06188
11. Pass, E.K., Cargile, P.A., DiTomasso, V., **Rodríguez Martínez, R.**, “Metallicities from High-Resolution TRES Spectra with The Payne and uberMS: Performance Benchmarks and Literature Comparison”, *Accepted to The Astrophysical Journal Supplement Series*. arXiv:2506.18961
10. Schulze, J.G., Wang, J., Johnson, J.A, (incl. **Rodríguez Martínez, R.**) “A Gap in the Densities of Small Planets Orbiting M Dwarfs: Rigorous Statistical Confirmation Using the Open-source Code RhoPop”, *The Planetary Science Journal*, Volume 5, Issue 3, id.71. arXiv:2403.13961
9. Armitage, T., Martin, D.V., **Rodríguez Martínez, R.**, “Identifying Flare Locations Through Exoplanet Transit Occultations”, *MNRAS*, 538, 4. arXiv:2501.04866
8. Boley, K.M., Panero, W.R., Unterborn, C.T., et al. (incl. **Rodríguez Martínez, R.**), “Fizzy Super-Earths: Impacts of Magma Composition on the Bulk Density of Lava Worlds”, *The Astrophysical Journal*, 954, 202. arXiv:2307.13726
7. Griffith, E.J., Johnson, J.A., Weinberg, D.H., et al., (incl. **Rodríguez Martínez, R.**), “Untangling the Sources of Abundance Dispersion in Low-Metallicity Stars”, *The Astrophysical Journal*, 944, 47. arXiv:2210.01821
6. Phillips, C., Wang, J., Edwards, B., **Rodríguez Martínez, R.**, et al. “Is LTT 1445 Ab a Hycean World or a cold Haber World? Exploring the Potential of *Twinkle* to Unveil Its Nature”, *MNRAS*, 526, 2. arXiv:2209.12919
5. Duck, A., Martin, D.V., Gill, S., et al. (incl. **Rodríguez Martínez, R.**) “The EBLM project X. Benchmark masses, radii and temperatures for two fully convective M-dwarfs using K2”, *MNRAS*, 521, 4. arXiv:2208.10534

4. Martin, D.V., Armitage, T., Duck, A., et al. (incl. **Rodríguez Martínez, R.**) “Revised Temperatures For Two Benchmark M-dwarfs – Outliers No More”, *under review*. arXiv:2208.10510
3. Fitzmaurice, E., Martin, D. V., **Rodríguez Martínez, R.**, et al. “Spectroscopy of TOI-1259B - an unpolluted white dwarf companion to an inflated warm Saturn”, *MNRAS*. arXiv:2206.01259
2. Godoy-Rivera, D., Tayar, J., Pinnsonneault, M., **Rodríguez Martínez, R.**, et al., “Testing the Limits of Precise Subgiant Characterization with APOGEE and Gaia: Opening a Window to Unprecedented Astrophysical Studies”, *The Astrophysical Journal*, 915, 19. arXiv:2101.09289
1. Rodriguez, J. E., Eastman, J. D., Zhou, G., et al. (incl. **Rodríguez Martínez, R.**) “KELT-24b: A $5M_J$ Planet on a 5.6 day Well-Aligned Orbit around the Young V=8.3 F-star HD 93148”, *The Astronomical Journal*, 158, 197. arXiv:1906.03276

AWARDS AND HONORS

Future Faculty Leaders Fellowship Center for Astrophysics Harvard & Smithsonian	September 2023
The Allan Markowitz Graduate Award in Observational Astronomy Department of Astronomy, Ohio State University	October 2022
ExoExplorers Program (ExoPAG Executive Committee/ NASA Exoplanet Exploration Program)	January 2021
Presidential Fellowship, The Ohio State University	December 2021
Two Sigma Diversity PhD Fellowship Grant (\$2000)	March 2021
Two Sigma Diversity PhD Fellowship Finalist	March 2021
Honorable Mention for Ford Foundation Predoctoral Fellowship	January 2019
Beth A. Brown Award for best graduate poster presentation (National Society of Black Physicists Conference)	November 2018
Ann S. Tuttle Scholarship (OSU)	September 2017

TALKS PRESENTED

“The Star-Planet Connection: Insights from Uniformly Characterized Exoplanet Systems” Tufts University, MA, Invited Colloquium Talk	April 24, 2026
“Characterizing Exoplanets and their Host Stars: Insights from Individual Systems and Planetary Populations” Louisiana State University, LA, Invited Colloquium Talk	September 11, 2025
“Characterizing Exoplanets and their Host Stars: Insights from Individual Systems and Planetary Populations” University of Florida, Gainesville, FL, Invited Colloquium Talk	April 3rd, 2025
“Exploring the Properties and Compositional Diversity of Small Exoplanets” Wesleyan University, Middletown, CT, Invited Colloquium Talk	April 17, 2024
“Revelando la Diversidad Composicional y la Demográfica de Exoplanetas Super-Tierra” Universidad de Puerto Rico, Río Piedras, PR, Invited Colloquium Talk	March 20, 2024
“Exploring the Properties and Composition of Small Exoplanets” Harvard Origins of Life Initiative, Cambridge, MA, Invited Talk	March 7, 2024
“A comparison of the composition of planets in single- and multi planet systems orbiting M-dwarfs” Center for Astrophysics Seminar, Cambridge, MA.	March 5, 2024

“Revealing the Composition and Demographics of Low-mass Exoplanets” Penn State Exoplanet Seminar, State College, PA, Invited Seminar Talk	October 9, 2023
“On the Compositional Links Between Exoplanets and their Host Stars” NASA ExoPAG 28th Meeting, San Antonio, TX, Invited Talk	October 1st, 2023
“A Comparison of the Composition of Planets in Single-planet and Multiplanet Systems Orbiting M dwarfs ” NASA Science Interest Group 2 (Demographics) Virtual Meeting, Invited Talk	September 13, 2023
“Journey into the Centers of Super-Earths: Revealing the Compositional Diversity of Small Exoplanets”, 241st Meeting of the American Astronomical Society	January 9, 2023
“Revisiting the Composition of K2-106b: an Ultra-dense, Ultra-short Period Exoplanet” NASA JPL Exoplanet Seminar, Invited Talk	April 25, 2022
“Journey into the Centers of Super-Earths: Constraining the Interior Structure and Composition of Low-Mass Exoplanets” Carnegie Observatories, Invited Colloquium Talk	March 18, 2022
“Revisiting the Composition of K2-106b: an Ultra-dense, Ultra-short Period Exoplanet” ExoExplorer Science Talks	March 18, 2022
“Analytic Estimates of the Achievable Precision on the Physical Properties of Transiting Exoplanets” Habitable Worlds 2021 Conference	February 2021
“Analytic estimates of the achievable precision of the properties of transiting exoplanets” National Society of Black Physicists Conference	November 2020
“Finding Flares with ASAS-SN” Cool Stars Conference 2020, Invited Talk	Cancelled due to COVID-19
“KELT-25b & KELT-26b: A hot Jupiter and a substellar companion transiting young A-stars observed by TESS”, Carnegie Observatories, Invited “Astronomy Tea” talk	July 2020
“Searching for Hot Planets Around Hot Stars with the KELT Survey”, Harvard-Smithsonian Center for Astrophysics, Stars & Planets Seminar, Invited Talk	October 2019
“Strange New Worlds”, Arecibo Observatory, Invited Seminar Talk	April 2017

TELESCOPE PROPOSALS

Large Binocular Telescope (LBT), “Characterizing the atmospheres of white dwarf companions to known exoplanets” (PI), 10 hours awarded, August 2022

Large Binocular Telescope (LBT) “Confirming the First Super Mercury? Exploring the Composition of K2-229b, an Ultra-Dense, Ultra-Short Period Exoplanet” (PI), 8 hours awarded, August 2020

TEACHING & ADVISING

[The Ohio State University](#)

Summer Undergraduate Research Program (SURP) at OSU

Tayt Armitage, astronomy undergraduate, Co-advisor: David V. Martin

May 2022 - present

Columbia High School Science Research Program

Advisor for Nicholas Pugliese, high school student in the Science Research Program

January 2021 - December 2021

SciAccess Zenith Mentorship Program

Informal mentor for Olivia Wilkerson, high school student in the Space Science

Mentoring Program at OSU

August - December 2020

Polaris Mentorship Program

Kevin Hoy and Hannah Parsons; astronomy undergraduates, Co-advisor: Alison Duck August 2019 - May 2020

University of Puerto Rico, Río Piedras

University of Puerto Rico, Río Piedras, Intermediate Physics Lab II

Graduate Teaching Assistant (TA)

January 2016

University of Puerto Rico, Río Piedras, Introduction to Physics Lab I

Graduate Teaching Assistant (TA)

August 2015

***A tiempo* Tutoring Center**

Math and Physics Tutor for Middle School students

August 2011

OUTREACH & SERVICE

Co-organizer for the CfA Staff Art Show event (Cambridge, MA)

May 2026

Reviewer for the NASA XRP Program

Referee for *The Astronomical Journal (AJ)*, *Astronomy & Astrophysics (A&A)*,
and the *Monthly Notices of the Royal Astronomical Society (MNRAS)*

2023 - Present

Reviewer, NASA TESS Guest Investigator Program, Cycle 8 (2025)

June 2025

Speaker for the *Spanish Observatory Night*

at the Center for Astrophysics | Harvard & Smithsonian

June 2025

Panelist for *Women in Space*

Q&A webinar for elementary school and high school students

May 2022

Panelist for *Galaxy Quest* Monthly Movie Nights - Science Fiction vs Science Fact

Q&A webinar, Astronomy Department, OSU

August 2021

American Astronomical Society Journal Author Series

Discussion of the paper “Analytic Estimates of the Achievable Precision

on the Physical Properties of Transiting Planets Using Purely Empirical Measurements”

May 2021

Speaker for “Magnifying Voices in Physics” event through

Womxn in Physics and Astronomy (WiPA) at UC Irvine

May 2021

Invited speaker at the Columbus School for Girls

April 2021

Invited speaker for middle and high school students at Colegio

Congregacion Mita (Virtual)

March 2021

Panelist for Space4Women Show (Webinar)

February 2021

Invited Speaker for OSU’s Astronomical Society

November 2020

Speaker for Friends of Ohio State Astronomy and Astrophysics (FOSAA) event

“The Search for Life in the Universe and the Exoplanet Revolution”

October 2020

Panelist for *Women in Space*

Q&A webinar for elementary school and high school students

September 2020

Panelist for *The Martian* Monthly Movie Nights - Science Fiction vs Science Fact

Q&A webinar, Astronomy Department, OSU

August 2020

Invited speaker for the Planetary Habitability Lab’s “AstroChat” series event

University of Puerto Rico Arecibo campus

December 2019

Invited speaker at elementary school Myrna M. Fuentes

Caguas, Puerto Rico

December 2019

Creator and co-organizer of symposium: “The Hunt for Alien Habitable Worlds”

Speakers: Sarah Ballard, Noemí Pinilla-Alonso, Abel Méndez

University of Puerto Rico, Río Piedras

March, 2017

PRESS/MEDIA COVERAGE

Astrobiters: Siblings or Only Child: M Dwarf Planets ([Link](#))

Reaching for the Stars: Interview for ¿Qué Pasa, Ohio State? magazine ([Link](#))

AAS Journal Author Series Interview ([Link](#))

Space.com Article ([Link](#))

Physics.org featured article on ASASS-SN-18di discovery ([Link](#))

POSTER PRESENTATIONS

“A Uniform Determination of the Bulk Metallicities and Alpha-Process Elemental Abundances of Confirmed Exoplanet Systems in the Northern Hemisphere”

Know Thy Star 2 Conference

February 2025

“Discovery and Characterization of an Eccentric Warm Saturn Transiting the Solar Analog TOI-4994”

TESS Conference

August 2024

“A Comparison of the Composition of Planets in Single-planet and Multiplanet Systems Orbiting M dwarfs”

GMT Conference

September 2023

“Is K2-106b truly a Super-Mercury?”, Emerging Researchers in Exoplanet Science Conference (ERES)

August 2022

“Is K2-106b truly a Super-Mercury?”, Exoplanets IV Conference

May 2022

“Finding Flares with ASAS-SN”, National Society of

Black Physicists Meeting, won Beth Brown Award for Best Graduate Poster

November 2018

“Finding Flares with ASAS-SN” Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) Conference

October 2018

“Characterization of Low-mass K2 planet hosts using Near-Infrared Spectroscopy”

229th Meeting of the American Astronomical Society

January 2017

“Characterization of Low-mass K2 planet hosts using Near-Infrared Spectroscopy”

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) Conference

October 2016

CONFERENCES ATTENDED

Exoplanets 6 (Porto, Portugal)

June 2026

Know Thy Star 2 (Pasadena, CA)

February 2025

TESS Science Conference III (Cambridge, MA)

August 2024

Exoplanets V (Leiden, Netherlands)

June 2024

241st Meeting of the American Astronomical Society (Seattle, WA)

January 2023

Great Lakes Exoplanet Area Meeting (GLEAM) (Columbus, OH)	November 2022
Emerging Researchers in Exoplanet Science (Pennsylvania State, PA)	August 2022
Exoplanets IV (Las Vegas, NV)	May 2022
Habitable Worlds 2021 (Virtual)	February 2021
Exoplanet Demographics Conference (Virtual)	November 2020
National Society of Black Physicists (Virtual)	November 2020
233rd Meeting of the American Astronomical Society (Seattle, WA)	January 2019
National Society of Black Physicists (Columbus, OH)	November 2018
SACNAS - National Diversity in STEM Conference (San Antonio, TX)	October 2018
Planets around Hot Stars Meeting (Nashville, TN)	June 2018
2nd Earth-Like Worlds Workshop (Manati, PR)	February 2017
229th Meeting of the American Astronomical Society (Grapevine, TX)	January 2017
SACNAS - National Diversity in STEM Conference (Long Beach, CA)	October 2016
Astrobiology Science Conference (Chicago, IL)	June 2015